Objects

JSON = JavaScript Object Notation. A function inside an object is called a **method**.

```
const product = {
  name: 'socks',
  price: 190
};
```

A basic object. **name** *is a* **property**, **'socks'** *is the* **value**.

```
console.log(product.name);
```

Access a property value (dot notation).

```
product.name = 'token';
```

Change a property value.

```
console.log(product);
```

Print the whole object.

```
product.newProperty = 45;
```

Add a new property.

```
delete product.newProperty;
```

Delete a property.

```
product['name'];
```

Access value using **bracket notation**.

```
const obj = {
  function1() {
    console.log('function inside obj');
  }
};
```

Functions inside objects are called **methods**.

Auto Boxing

When you call a method on a **primitive**, *JS temporarily wraps it in an object, uses the method, then throws the wrapper away.*

```
"hello".toUpperCase()
```

JS creates a temporary string object to call the method.

Objects are References

Reference Assignment

Assigning an object to a variable copies the **reference** *(memory address), not the actual value.*

Object Shortcuts

1. Destructuring

Instead of `const name = obj.name;`

```
const { name } = obj;
```

*Extracts the **name** property into a variable named **name**.*

2. Shorthand Property

```
const obj = { name };
```

Instead of `const obj = { name: name };`

If variable name matches property name, you can write it once.

3. Shorthand Method

```
const obj = {
  greet() { console.log("hi"); }
};
```

Instead of `greet: function() { ... }`

*The **function** keyword is optional inside an object.*

JSON

⚠ **null** *being* **typeof object** *is a bug in JS, not a feature.*

```
const jsonStr = JSON.stringify(obj);
```

Converts JS Object → **plain JSON string**.

```
const obj = JSON.parse(jsonStr);
```

Converts JSON string → **JS Object**.

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localStorage

localStorage only saves **strings** *— use JSON.stringify/parse for objects.*

```
localStorage.setItem("Name", "Vishwanath");
```

Store an item.

```
localStorage.getItem("Name");
```

Get an item.

```
localStorage.removeItem("Name");
```

Remove a specific item.

```
localStorage.clear();
```

Clear everything in localStorage.

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null vs undefined

null

Intentional absence *("I want* **here** *to be empty").*

undefined

JS says it exists but hasn't been **declared/assigned** *yet.*